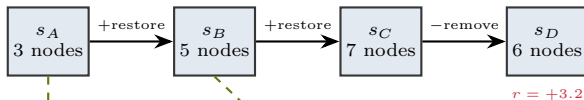


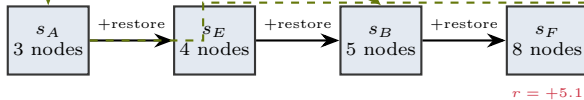
How We Think Q-Learning Works

Idealized: states generalize, Q-values transfer across episodes

Episode 1



Episode 2



Q-Table

$(s_A, restore\ 42) = 0.8$	{ Ep 1
$(s_B, restore\ 17) = 1.2$	
$(s_A, restore\ 99) = 0.3$	
$(s_B, remove\ 42) = -0.4$	{ Ep 2 update
$(s_A, restore\ 42) = 1.4$	
$(s_C, remove\ 17) = 0.6$	

Revisiting s_A in Episode 2 refines $Q(s_A, \cdot)$ with new experience.
Over 30 episodes, frequently visited states accumulate reliable Q-values → **convergence**.